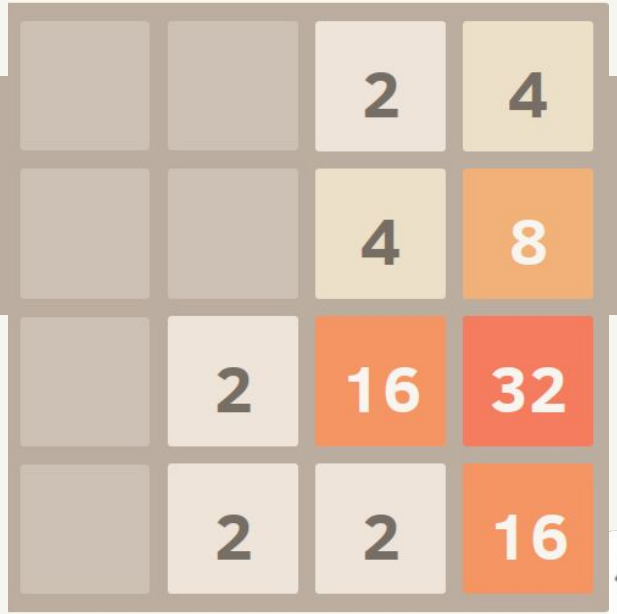
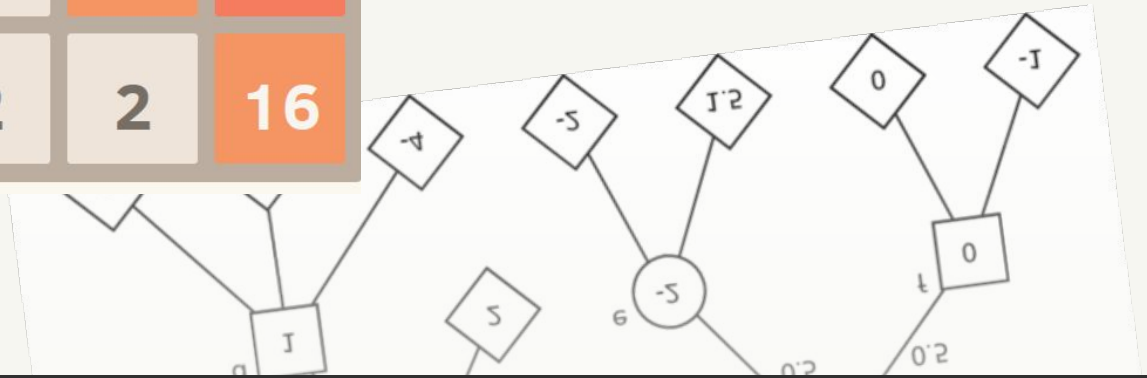


Beating 2048 With Expectimax

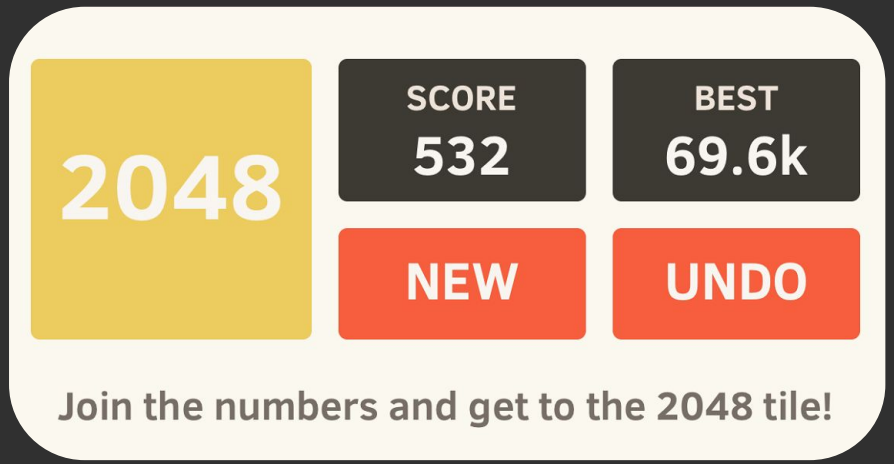


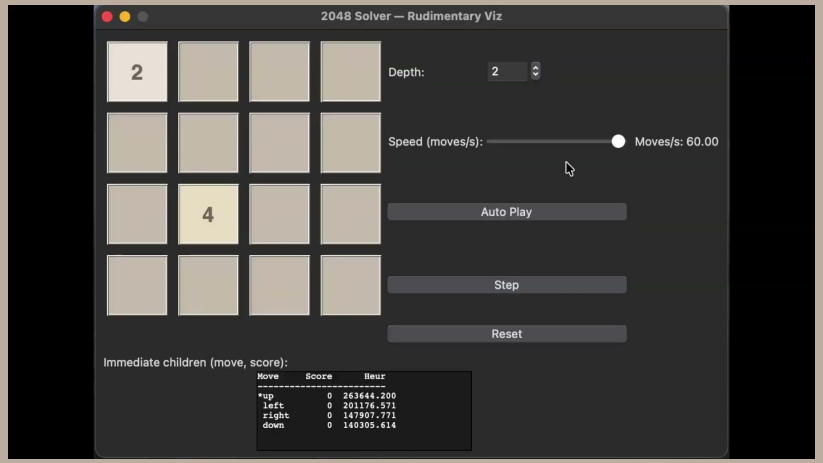
Christian Jahnel, Akil Dickerson, Douglas Parker, Sang Woo, Philip Roh

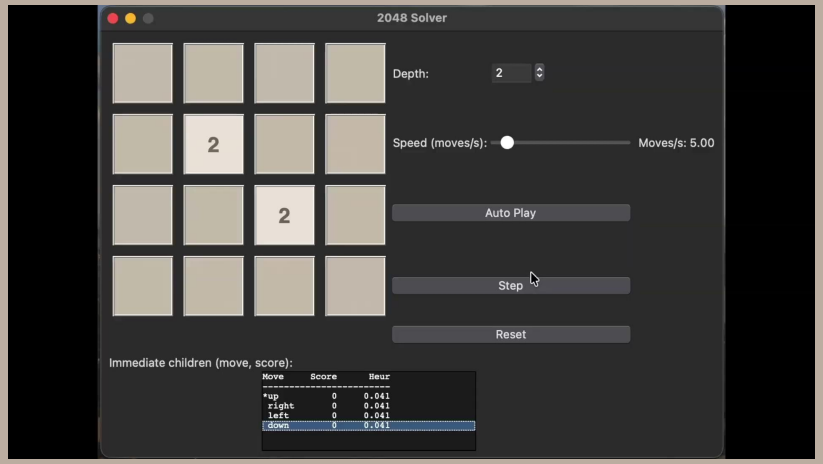
COMP 560

2048 is a single-player sliding tile puzzle game where the objective is to slide numbered tiles on a grid to combine them, **aiming to create a tile with the value 2048.** The game requires strategic movement to prevent the grid from filling up.

State:	{4x4 grid of cells, Score}
Succ. Function:	{Up, Left, Down, Right}
Goal State:	any cell == 2048

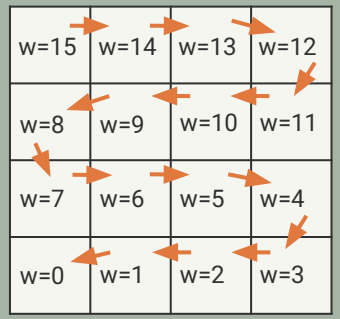






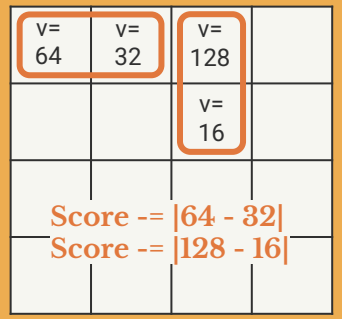
Snake Heuristic

Multiply a tile value by that cell's assigned weight, and add them all up. Weights are initialized in a descending "snake" pattern which **rewards having a descending chain of large values in the corner.**



Smooth Heuristic

Take every pair of tiles that are touching and calculate the absolute difference between them and subtract that from a running score. **How easily can these tiles merge together right now?**



Empty Heuristic

Count the number of empty squares on the 4x4 grid and normalize this count by dividing by 16. This represents future flexibility. **Not playing to win, but to not lose on the next turn.**

Comb. Heuristic

Normalize all three heuristics into a standard range (usually 0 to 1). Then, add them all together. **Acts like a committee between all the heuristics.**

Heuristic	Depth	Avg	Max	Min	Std	MaxTile	≥2048	≥1024
snake	1	3,299	10,440	756	1,948	1024	0/25	1/25
snake	2	3,208	7,140	740	1,562	512	0/25	0/25
snake*	**3**	**35,599**	**111,280**	**3,156**	**22,195**	**8192**	**20/25**	**22/25**
empty	1	2,990	6,724	788	1,356	512	0/25	0/25
empty	2	2,990	6,724	788	1,356	512	0/25	0/25
empty	3	9,075	15,568	3,352	3,792	1024	0/25	10/25
smooth	1	1,974	4,724	552	987	512	0/25	0/25
smooth	2	2,579	7,012	660	1,502	512	0/25	0/25
smooth	3	6,633	15,980	1,740	3,909	1024	0/25	5/25
combined	1	4,570	9,084	1,992	2,030	512	0/25	0/25
combined	2	6,296	14,368	1,848	3,170	1024	0/25	3/25
combined	3	18,220	58,688	5,448	12,095	4096	7/25	19/25

Human vs. AI Performance

Human Performance

Score	Max Tile
2776	256
10780	1024
3920	256
6792	512
1556	128
1224	128
1664	128
3200	256
14032	1024
6488	512
16944	1024
6268	512

Average Score: 6304
Median Max Tile: 384

AI Performance

Avg Score	Max Tile	
3299	1024	snake
3208	512	
35599	8192	
2990	512	empty
2990	512	
9075	1024	
1974	512	smooth
2579	512	
6633	1024	
4570	512	combined
6296	1024	
18220	4096	

Average Score: 8119*
Median Max Tile: 768*

*different heuristics/depths encompassed in combined statistic

Analysis

Clearly, the models outperformed everyone in the group. But, we are unable to extrapolate this Human-AI performance comparison to the highest human performers as we couldn't locate reliable data. With that said, this comparison still illustrates the difference in performance between the average human player of 2048 and our different heuristic-based AI models.